

The 9,450-hectare New Clark City development is physically located within the municipalities of Capas and Bamban in the province of Tarlac. Therefore, any data engineering pipeline must target Capas, Bamban, and Mabalacat (Pampanga) to capture the complete regional economic and transit corridor.

1. Road Networks & Transport Topology (Edge Graphs)

The simulation requires a topological graph of the physical roads, including the Subic-Clark-Tarlac Expressway (SCTEX), MacArthur Highway, and the newly constructed BCDA access roads.

- **Geofabrik OpenStreetMap (OSM) Extracts (Philippines):** The foundational road graph, it provides .osm.pbf and shapefiles containing lane counts, speed limits, and road classifications.

Link: <https://download.geofabrik.de/asia/philippines.html>

- **Humanitarian Data Exchange (HDX) - Philippines Road Surface Data:** Contains deep-learning derived classifications of paved versus unpaved roads (approximately 0.496 million km mapped), which is critical for modeling vehicular friction and speed limits in the peri-urban areas of Capas.

Link: <https://data.humdata.org/dataset/philippines-road-surface-data>

2. Built Environment, Cadastral, and Footprint Data

Because New Clark City is a planned development actively under construction, building footprints must be extracted from machine-learning generated polygons to render the 3D client and the obstacle configurations.

- **Google Open Buildings (v3):** High-resolution building footprints generated by Google AI from satellite imagery. Researchers can download the specific CSV/GeoJSON tiles covering the Central Luzon grid.
- **Microsoft Planetary Computer - Building Footprints:** Bing Maps ML-generated footprints formatted in Geoparquet. Highly accurate for identifying recently erected structures in the Clark corridor.

Link: <https://sites.research.google/open-buildings/>

- **BCDA Master Development Plan (FOI Portal):** The Bases Conversion and Development Authority (BCDA) holds the official zoning shapefiles for NCC (e.g., National Government Administrative Center, sports complexes, mixed-use zones). These spatial vectors must be requested via the Freedom of Information portal.

Link: <https://www.foi.gov.ph/agencies/bcda/>

3. Demographics & Synthetic Population Generators

To model the economic and social dimensions, the system requires granular population data to synthesize realistic commuter agents.

- **WorldPop Philippines (100m Resolution):** Provides GeoTIFF rasters estimating population counts and age-sex structures per 100-meter grid square. This acts as the baseline for Gravity Model trip generation without violating data privacy.
Link: <https://hub.worldpop.org/geodata/summary?id=6316>
- **HDX Philippines Population Projections by Admin3:** Contains population projections (2020 to 2025) by city and municipality (Admin3 level), based on the Philippine Statistics Authority (PSA) Census data.
Link: <https://data.humdata.org/dataset/philippines-population-projection-2020-2025-admin3>
- **PSA OpenSTAT Database:** The official repository for tabular Community-Based Monitoring System (CBMS) and Census of Population and Housing (CPH) data for Tarlac and Pampanga.
Link: <https://openstat.psa.gov.ph>
- **Kaggle: Philippines Population Projection Dataset:** A formatted subset of the HDX projection data optimized for machine learning workflows and synthetic population sampling.
Link: <https://www.kaggle.com/datasets/jocelyndumlao/philippines-population-projection-2020-to-2025>

4. Hazard Vectors, Terrain Modeling, and Resilience Rasters

To evaluate infrastructure under extreme-event resilience testing, terrain and hazard data are required to dynamically invalidate roads during simulations.

- **GeoRiskPH / HazardHunterPH (DOST-PHIVOLCS):** The authoritative repository for fault line traces, seismic buffers, and liquefaction susceptibility polygons for Central Luzon.
Link: <https://hazardhunter.georisk.gov.ph>
- **UP NOAH (Nationwide Operational Assessment of Hazards):** Provides raster inundation maps for 5-year, 25-year, and 100-year return period floods, critical for simulating monsoon shocks.
Link: <https://noah.up.edu.ph>
- **Copernicus DEM GLO-30 (via OpenTopography):** A global 30-meter resolution Digital Elevation Model. To calculate road slopes, vehicular drag, and localized flood pooling across the Tarlac terrain.

Link: <https://opentopography.org>

5. Traffic, Mobility, and Economic Baselines (Kaggle & Gov Data)

To provide the empirical ground-truth for the glass-box traceability and validation modules, researchers can leverage specific telemetry and economic datasets.

- **Kaggle: Techsalerator Foot Traffic & Mobility Data for Philippines:** A dataset capturing location visitation trends, origin-destination mobility patterns, and peak hour congestion metrics. This is highly useful for calibrating the temporal distribution curves of agents.

Link:

<https://www.kaggle.com/datasets/techsalerator/foot-traffic-and-mobility-data-for-philippines>

- **Kaggle: Smart Mobility Traffic Dataset:** Contains features such as vehicle counts, speeds, road occupancy, and environmental factors designed to support machine learning models in predicting traffic congestion levels.

Link: <https://www.kaggle.com/datasets/ziya07/smart-mobility-traffic-dataset>

- **Bureau of Internal Revenue (BIR) Zonal Values Portal:** To calculate the economic impact dimension and land value uplift, planners must download the official Schedule of Zonal Values for RDO No. 17 (Tarlac City/Capas/Bamban) and RDO No. 21B (South Pampanga).

Link: <https://www.bir.gov.ph/index.php/zonal-values.html>